BALL STATE UNIVERSITY

Module 2 Lecture - Descriptive Statistics and Plots

Introduction to Statistical Methods

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"I don't mind not knowing. It doesn't scare me." — Richard P. Feynman

1 Overview and Introduction

1.1 Objectives

- Display data graphically and interpret graphs
- Recognize, describe, and calculate the measures of location of data (quantiles and percentiles)
- Recognize, describe, and calculate the measures of center tendency
- Recognize, describe, and calculate the measures of dispersion
- Understand the complementary value of visual and numeric descriptions of data
- Be able to compare and contrast the use cases for different metrics

1.2 Introduction

- Data, in it's raw form, is very ____________
  - Especially large data sets are likely to be borderline ____________

Discuss: With the following table, try to say something meaningful about this data/explain it

Student	Q1	Q2	Q3	Q4	Q5
Student 1	1	0	1	1	0
Student 2	0	1	0	1	1
Student 3	1	1	1	0	0
Student 4	0	0	1	0	1
Student 5	1	1	0	1	1

- Realistically, we need a way to ____________ our datasets in a way that capture the overarching trends in the values
  - This can be done both ____________ and via ____________ statistics, and often times, both!
  - Though this is a statistics class, sometimes the graphs are the most understandable part of a research paper!

Important

There is usually not only one 'right' way to graph or represent data - it may be advantageous to try multiple methods and see how they compare

- The book is filled with examples and practices to get better at navigating these, please try some of them!

2 Descriptive Statistics

2.1 Percentiles and Quartiles

- Quartiles are used to cut numerical data into 4 equal-sized ______ when the data is ordered smallest to largest
  - Q_1 (first quartile) is above 1/4 or 25% of the data
  - Q_2 (second quartile) is above 1/2 or 50% of the data, or the median
  - Q_3 (third quartile) is above 3/4 or 75% of the data
  - For example, if I state $Q_3 = 20$, that means that the value of 20 is above 75% of the data
  - Used in Box plots, we'll get to that in a bit

Discuss: If quartiles split the data into 4 equal sections, how many sections are in quantiles

- Percentiles function the same as quartiles, but ______ the data into 100 equal-sized sections
  - For example, the at the 90th percentile, 90% of scores are lower
  - Percentiles are especially commonly used to describe roughly where ______ data points fall - this comes up often in standardized testing
  - The 25th percentile is equivalent to Q_1, 50th percentile is equivalent to Q_2 (and thus, the median), and the 75th percentile is equivalent to Q_3
  - For example, if I state that the 23rd percentile is 45, then the value of 45 is above 23% of the data

2.1.1 A Formula for Finding the Kth Percentile

- Often times, we want to know what ________________ corresponds to a certain percentile in a dataset
  - We can use the following formula to calculate this

$$i = \frac{k}{100} * (n + 1)$$

- Where:
  - i is the index or rank of the value when ordered smallest to largest
  - k is the k^{th} percentile
  - n is the total number of data points
- If i ends up being between two integers, i.e., a decimal, then we ________________ up and down to the nearest ranks and take the average of their integers
  - It's worth mentioning this is just one of ________________ procedures for finding the k^{th} percentile…
- For a formula like this, it is easiest to approach it algebraically, just inserting the chosen values over the letters in the formula

> Discuss: Try writing out this same equation, replacing k with 20 and n with 12 - could you solve it from here?

2.1.2 Example of Finding Kth Percentile

Dataset: 5, 6, 7, 8, 9 (ordered smallest to largest)

Prompt: What value corresponds to the 70th percentile?

$$i = \frac{70}{100} * (5 + 1)$$

$$i = 0.70 * 6$$

$$i = 4.2$$

We round i up and down to the 4th and 5th ranks, which correspond to values 8 and 9 in the dataset, their average is 8.5 $\longrightarrow$ 70th percentile

? If I wanted to find what value corresponds to Q_2 what k would I use in this formula?

- (●) A) 10
- ( ) B) 20
- ( ) C) 25
- ( ) D) 50

Explanation:

2.1.3 A Formula for Finding the Percentile of a Value in a Data Set

- We can go the ________________ direction to figure out what a specific datapoint's percentile is

$$\frac{x + 0.5y}{n} * 100$$

- Where:
  - x is the number of data points up to and NOT including the point of interest
  - y the number of occurrences of the values of interest
  - n is the total number of data points
- Like the prior formula, this is just *one* procedure, you may run into others in the wild

2.1.4 Example of Finding Percentile of a Value in a Data Set

Dataset: 5, 6, 7, 8, 9 (ordered smallest to largest)

Prompt: What percentile is value '8' at?

$$\frac{3 + 0.5(1)}{5} * 100$$

$$\frac{3.5}{5} * 100$$

$$70 \rightarrow percentile$$

- **WAIT!** Didn't we just say that 8.5 is the 70th percentile in the last example?
  - Yes, that is because these formulas are only ________________, and work poorly in small datasets
  - There are several slight variations on calculating these, but the formulas above is what we will use for this class

2.2 Descriptive Tables and Frequencies

- Simple frequency is a count of the ________________ of cases – or frequency - that fall in the different ________________ or that obtain certain scores.
  - E.g., - the number of girls and boys in a sample the number of students who identify themselves as African-American, Asian- American, Hispanic, or White on a questionnaire; the number of people who obtain different scores on a test
- Relative frequency is the ________________ or percentage of cases for each score in the distribution
- Cumulative frequency is a count of the number of cases that fall ________________ a certain score. A cumulative frequency is provided for all scores in the distribution
- Cumulative relative frequency is the ________________ or percentage of cases that fall below a certain score.

Score	Frequency	Cumulative Frequency	Percent	Valid Percent	Cumulative Percent
0	1	1	2.2	2.2	2.2
2	1	2	2.2	2.2	4.3
3	3	5	6.5	6.5	10.9
4	2	7	4.3	4.3	15.2
5	4	11	8.7	8.7	23.9
6	4	15	8.7	8.7	32.6
7	6	21	13.0	13.0	45.7
8	3	24	6.5	6.5	52.2
9	7	31	15.2	15.2	67.4
10	7	38	15.2	15.2	82.6
11	3	41	6.5	6.5	89.1
12	2	43	4.3	4.3	93.5
13	2	45	4.3	4.3	97.8
15	1	46	2.2	2.2	100.0
Total	46		100.0	100.0	

Discuss: Based on your reading of the above table, what is the cumulative frequency at score '11'?

2.3 Measures of Central Tendency

- Measures of central tendency of a ________ are statistics that ________ scores that are typical, central or average in the distribution.
  - The most ________ measures of central tendency in the social sciences are the mode, the median, and the mean.
- The mode is the most commonly ________ value in a variable/dataset
  - Mode is often just spelled out, and not given a special notation
  - There is not a very useful formula for finding the mode, but realistically all you need to do use Bar Graphs and find the ________ bar (for the highest frequency)

- Example of finding mode of a sample dataset:
  - I have 5 students with the following sexes: $\{F, M, F, F, M\}$
  - I have 3 female students, and 2 male students, thus the **sample mode of the sex variable is 'Female'**, as that is the most common value in that variable
- The median is the value at the 50th percentile, second quartile (Q_2), or, put simply, the value that ________ the data into two equal halves when ordered from smallest to largest
  - In the case that there is no actual exact middle value when ordered smallest to largest, we'd ________ the two middle-most values
  - Sample median ________ notation: X_m
  - Population median ________ notation: M
  - The best strategy for finding median is to order the values and then cross values off from both ends until you end up with one uncrossed value
  - To find the *location* of the median, one can do: $\frac{n+1}{2}$, where:
    - n: number of observations/people
- Example of finding median of a sample dataset:
  - I have the following test scores: $\{23, 61, 35, 89, 74\}$
  - First, we have to order the data smallest to largest: $\{23, 35, 61, 74, 89\}$
  - If I cross off each side of the data: $\{\cancel{23}, \cancel{35}, 61, \cancel{74}, \cancel{89}\}$
  - Thus, my **sample median of test scores is 61** or $X_{M,TestScore} = 61$
  - Using the location formula: $\frac{5+1}{2} = 3 \rightarrow$ the 3rd number in the order is the median
- The mean is the arithmetic average of the data
  - Sample mean ________ notation: $\bar{x}$ (x-bar)
  - Population mean ________ notation: μ (mu)
  - The sample mean is calculated with:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + ... + x_n}{n}$$

- Example of finding mean of a sample dataset
  - I have the following GPAs: $\{1.00, 1.00, 2.50, 3.00, 4.00\}$
  - Using the formula: $\frac{1+1+2.50+3.00+4.00}{n} = \frac{11.50}{5} = 2.3$
  - Thus, **my sample mean of GPAs is 2.3** or $\bar{x}_{GPA} = 2.3$

Important

Something worth noting is that the median is less prone to be affected by outliers than the mean, making it useful in certain circumstances.

? I have data 1,2,2,2,3,4,5. Datapoint 2 is the most commonly occurring value, thus it is the [BLANK]

- (●) A) Mode
- ( ) B) Median
- ( ) C) Mean
- ( ) D) None of the above

Explanation:

Discuss: Find the mean, median, and mode of the following data: 1, 1, 2, 3, 5, 6, 8, 9, 10

2.4 The Law of Large Numbers and the Mean

- As a general rule, as the sample size (n) grows, $\bar{x}$ and μ converge - this is true of most statistic - parameter relationships
  - This is related to the central limit theorem, to be discussed slightly later
  - However, this also explains why bigger samples are often treated as ____________ in a lot of scientific research - because the sample statistics better approximate the parameters

Discuss: Review: Try explaining, from Module 1, why a more representative statistic is a good thing for our research?

2.5 Measures of Dispersion

- Measures of dispersion are statistics that summarize how data is ____________ out across a distribution
  - The most ____________ measures of dispersion we should be concerned with are standard deviation, variance, range, and interquartile range.
- Each number in a dataset has a deviation, or how ____________ away it is from the mean
  - This is calculated simply as the ____________ between the value and the sample mean of the data
  - $x - \bar{x}$ or
  - $x_i - \bar{x}$
- Example of calculating deviation of a single point:
  - We have dataset with $\bar{x} = 5$, and we are interested in the individual point of 10
  - Using the deviation formula: $10 - 5 = 5$, thus the deviation of point 10 is 5
- To calculate the standard deviation, we first will calculate the variance, which is just the ____________ standard deviation
  - Sample variance statistic: s^2
  - Population variance parameter: σ^2
  - The variance sample statistic and population parameter formulas are, respectively:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma^2 = \frac{\sum (x - \mu)^2}{n}$$

> **Important**
>
> The reason we must use variance to calculate standard deviation relates to the need to square in order to avoid getting 0 from summing the deviations

- By far, the most important and most used way to describe the ____________ of all the data is the standard deviation
  - A measure of the average ____________ away from the mean that a point is in a distribution

  - It is used instead of variance because it removes the "square" from interpretation
  - Sample standard deviation statistic: s
  - Population standard deviation parameter: σ (sigma)
  - The standard deviation sample statistic and population parameter formulas are, respectively:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{n}}$$

> **Important**
>
> Realistically, we tend to only directly use the sample statistics calculations

> Discuss: Look back at the previous equations; which of the measures of central tendency sees the greatest use in these formulas?

- The range is the ______ between the largest and smallest values in the data
- Example of calculating range in a sample:
  - Data: $\{4, 2, 6, 8, 20\}$
  - Subtracting smallest and largest values: $20 - 2 = 18$, **18 is the range of the sample**
- The interquartile range (IQR) is the 75th percentile (upper quartile, Q_3) minus the 25th percentile (lower quartile, Q_1). It is the width of the interval that contains the ______ 50% of the data.
  - In formula it could be represented as: $Q_3 - Q_1 = IQR$

2.6 Calculating Measures of Dispersion

2.6.1 Calculating Variance and Standard Deviation

While not necessary with computers and calculators, it can be useful to work out statistics "by hand" for learning how they work. For example:

Dataset: 1, 2, 3, 4, 5

Size of sample: $n = 5$

Sample Mean:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + ... + x_n}{n}$$

$$3 = \frac{1 + 2 + 3 + 4 + 5}{5}$$

Sample Variance:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Hint: whenever you see a $\sum$ sign, follow this tabular procedure

x	x - xbar	(x-xbar)^2
1	1 - 3	-2^2
2	2 - 3	-1^2
3	3 - 3	0^2
4	4 - 3	1^2
5	5 - 3	2^2
	Sum	10

$$2.5 = \frac{10}{5 - 1}$$

Sample Standard Deviation:

$$s = \sqrt{s^2}$$

$$1.58 = \sqrt{2.5}$$

2.6.2 Describing Points and Values with Standard Deviations (Z-scores)

- Much like describing a specific value with a percentile, we may want to describe how far away from the ______ a certain point is, and we can do so using the standard deviation
- We can do this with what are called z-score with the following formula for sample:

$$z = \frac{x - \bar{x}}{s}$$

- Where: x is a individual data value of interest s is the standard deviation

By hand (pulling from last example):

$$-0.63 = \frac{2 - 3}{1.58}$$

- We could then say that the value of 2 is -0.63 standard deviations away from the mean of 3

> Discuss: Try working out the z-score of 5 given that same dataset

- Z-scores are often used in conjunction with Percentiles and Quartiles, to describe a position of an individual in a broader dataset

2.7 Aside on Outliers

- An outlier, also known as an **extreme value**, is a data point that falls ______ from the others, somehow breaking from the pattern of the data
  - While more commonly applied to numeric data, it could sometimes be used to describe a single member of a qualitative ______
  - It is characteristic for this point to have an extreme z-score, i.e., be many standard deviations away from the ______

> **Important**
>
> 'Outlier' can be a frustratingly loose term in statistics, because there is not one always accepted answer for just how far a point needs to be removed to be an outlier. Make sure to read carefully to see how any one given author defines it in an individual example or study.

- Many graphical methods naturally ______ outliers in the data
  - In plots, you are mainly just looking for visual ______, or breaks in the visual pattern

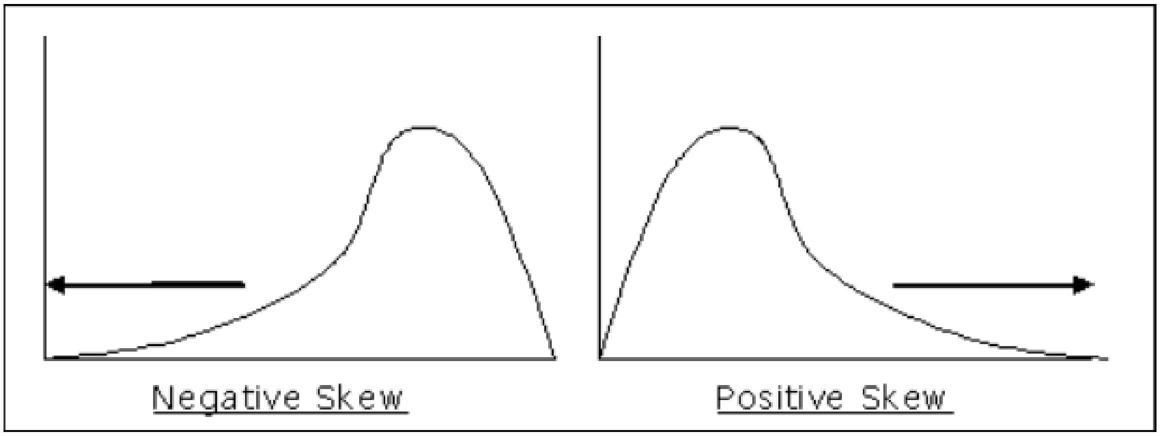
2.8 Descriptions of Distributions

- There are several ______ that are not well described as either Measures of Dispersion or Measures of Central Tendency, but still helpful in understanding the layout of a continuous distribution
- Usually, we use The Normal Distribution as an "ideal", and describe variables' distribution in how it ______ to the normal distribution
  - This is how we will describe Skewness, Kurtosis, and Modality
- There are several different ways to describe these ______ with numeric statistics, but for now, we'll just focus on how they appear graphically in a histogram

2.8.1 Skewness

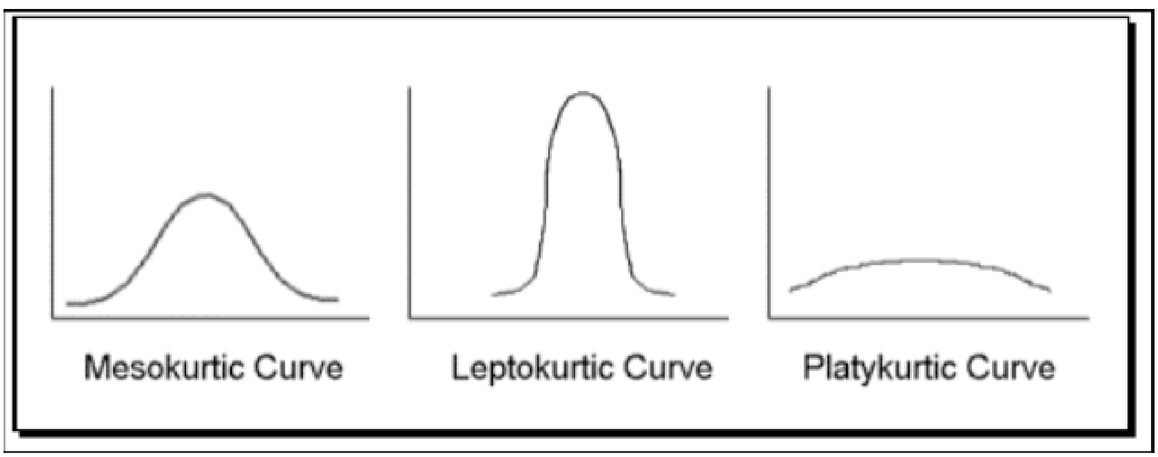
- Skewness is a description of how a distribution departs from ______ or has a 'tail'
  - A positive/right skew: when a distribution has most values clustered towards the ______ end, and a tail out to the right side of the frequency histogram
  - A negative/left skew: when a distribution has most values clustered towards the ______ end, and a tail out to the left side of the frequency histogram

> **Important**
>
> As a helpful mnemonic, the direction of the skew is the direction of the 'tail', not the 'hump'



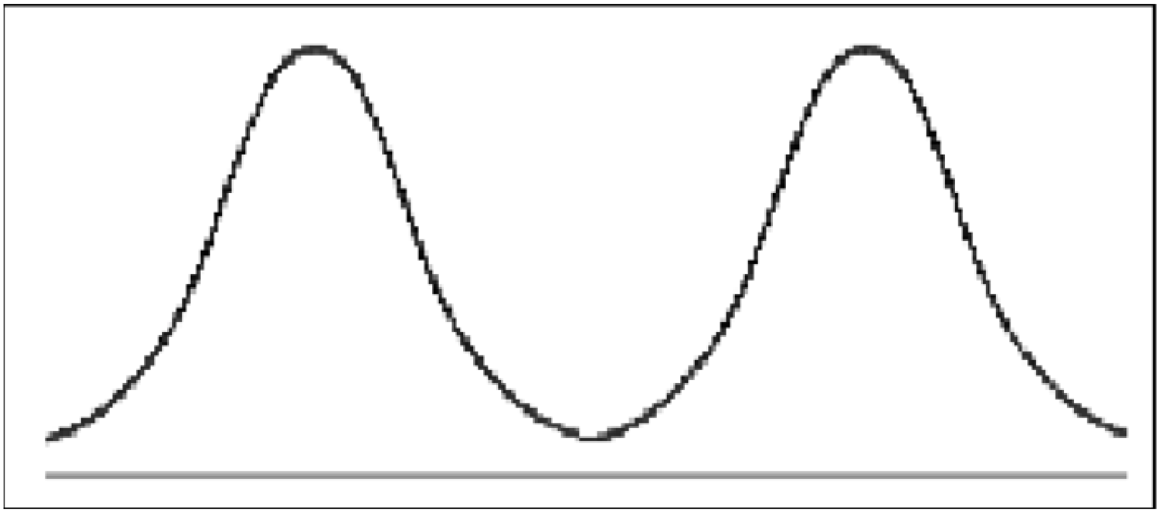
2.8.2 Kurtosis

- Kurtosis is a description of how 'peaked' a distribution is
  - A leptokurtic distribution has too many scores concentrated in the center and not ______________ in the tails. (peaked)
  - A platykurtic distribution has too few scores in the center and too many in the ______________. (flat)
  - The Normal Distribution is said to be ______________



2.8.3 Modality

- Modality a description of how many ______________ a distribution has
  - A distribution is unimodal when is has only one peak, and bimodal when it has two peaks

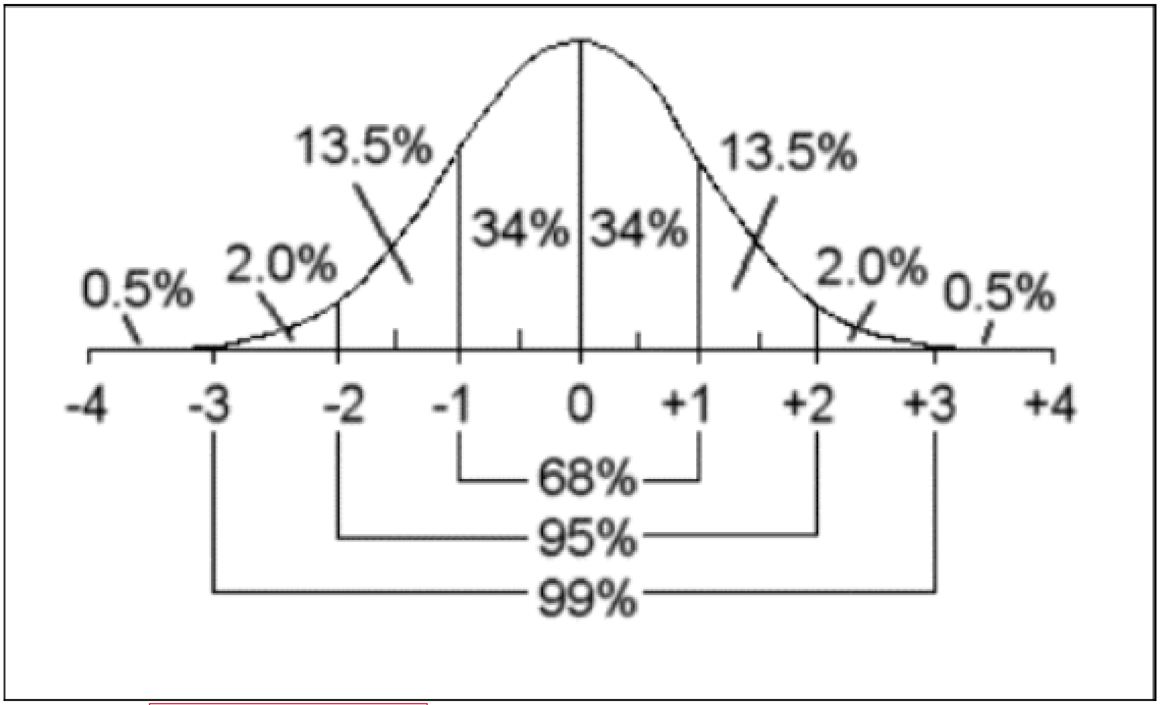


Discuss: Take a guess at what a distribution with three peaks would be called

2.8.4 The Normal Distribution

Important

The normal distribution, is an 'ideal' and used theoretically. In practice, collected raw data will never be perfectly normal. There are also other ideal distributions such as the exponential and uniform distributions, that are also useful in certain circumstances.



- The normal distribution is a distribution of a ________ variable that is unimodal, symmetrical, mesokurtic, bell-shaped, and the mean, median, and mode are equal
  - Follows the empirical rule
  - We'll revisit this in it's own module later on

Discuss: Reading the distribution chart above, at what number of standard deviations are 84 percent of values beneath it?

2.9 Sampling Variability of Statistics

- As mentioned prior, different ________ from the same population of interest will not be exactly the same, described as sampling variation
  - Thus, their data, means, standard deviations will all be somewhat different every time we take a new sample
  - If we were to take many ________ samples, we would see slight variation across all of them

- Put together, they are all just different ________ of their parameters (but the parameter is what we are after!)
- Review:
  - $\bar{x}$ estimates μ
  - X_M estimates M
  - s estimates σ

3 Descriptive Plots

3.1 Introduction

- In this first section, we looked at primarily statistics and numeric representations, now we'll look at some graphs used often to show ________ of certain values in the data

? Review: Which of the following is NOT quantitative data?

- (●) A) Hair color
- ( ) B) Age (in years)
- ( ) C) Height (in cms)
- ( ) D) Total test score (in number of points)

Explanation:

3.2 Stem-and-Leaf Plots

- **Stem-and-leaf graphs**, also known as **stemplots**, are technically a ________ method to represent data - but function to visually inspect the distribution of the data.
- Stemplots a valid choice for representing ________, quantitative data for a single variable
  - However it works ________ if the range of values is reasonably restricted, and with the same decimal structure
  - Continuous data ________ work, but discrete data can be somewhat easier
  - E.g., Test scores ranging from 0 - 100 $\rightarrow$ ________!

- E.g., Reaction time ranging from 0.50 secs - 420 seconds $\longrightarrow$ ________ !

- Stemplots contain a **leaf** which contains the **final significant digit** (in practice this is usually just the ________ digit)
  - Then, the ________ is everything *but*, the digits of the leaf
  - Each number entry in a leaf represents a different value in the data
  - So technically, the number of digits that are leaves in the ________ of points in the data

- Practically, stemplots are ________ showing distribution, skew, and frequency of values
  - Especially common in ________
  - However, I do find they are ________ to interpret for non-statistical audiences

- Example of test scores 0 - 100: 2, 3, 5, 6, 9, 11, 14, 15, 16, 20, 21, 23, 27, 30, 32, 38, 41, 44, 46, 53, 55, 59, 60, 62, 67, 74, 79, 81, 85, 90

Stem	Leaf
0	2 3 5 6 9
1	1 4 5 6
2	0 1 3 7
3	0 2 8
4	1 4 6
5	3 5 9
6	0 2 7
7	4 9
8	1 5
9	0

Discuss: Try to explain why a stem-and-leaf-plot would NOT work for qualitative, nominal data

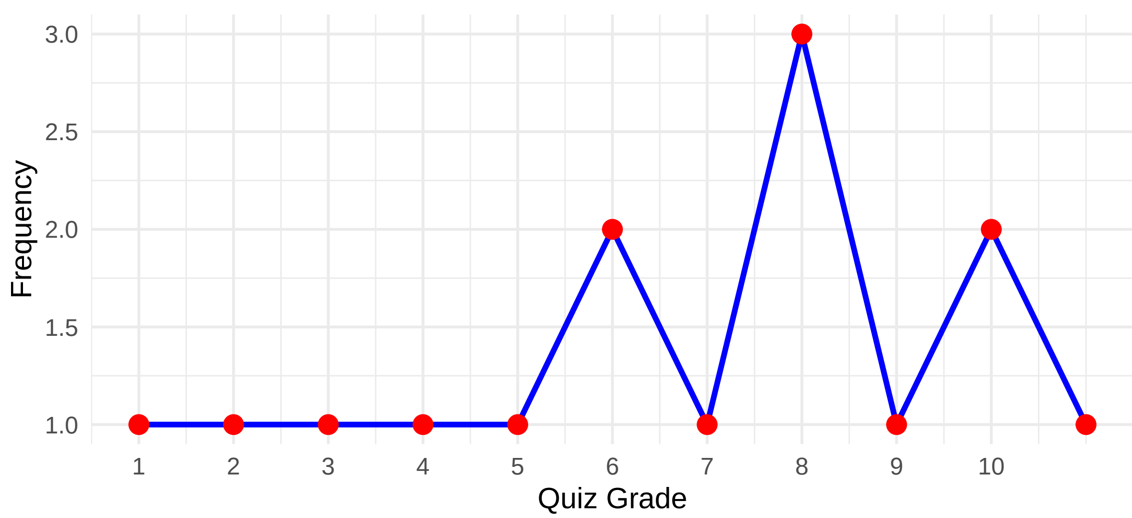
- As you'll see in the following line and bar graphs, one difference is that stemplots don't actually aggregate or summaries the data at all
  - One nice thing about these plots is that they actually leave the numbers somewhat ________, so you can see the entire dataset at a

glance

3.3 Line Graphs

- **Line graphs** are a broad family of plots that always have some dotted data points ______________ by a continuous line
  - For our purpose, they are another way to visualize ______________ of data points
- Unlike stemplots, they could technically be used for qualitative data (but I wouldn't recommend that use)
  - Otherwise, the same rules for ______________ apply from the stem-plots
- In the line plot:
  - The height, or **y-axis**, represents the ______________ of a certain value
  - The place on the **x-axis** represents what value's frequency is ______________ by the y-axis

Figure 1: Frequency of Quiz Grades (0-10)



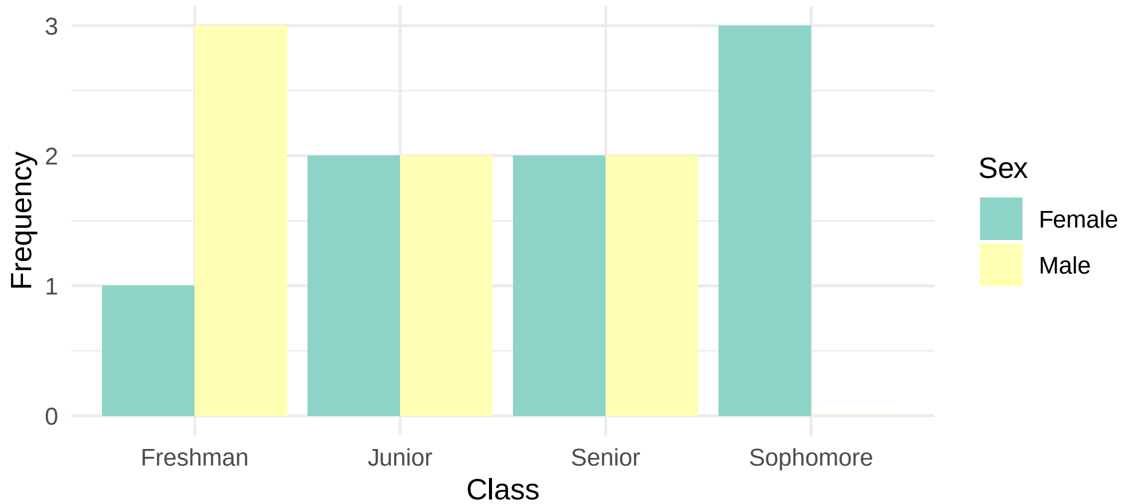
Discuss: Consider the above plot, do you feel there is anything confusing about interpreting it?

3.4 Bar Graphs

- Bar graphs help show ________________ of values of categorical/nominal variables

Figure 2: Frequency of Students by Class

Figure 3: Frequency of Students by Class and Sex



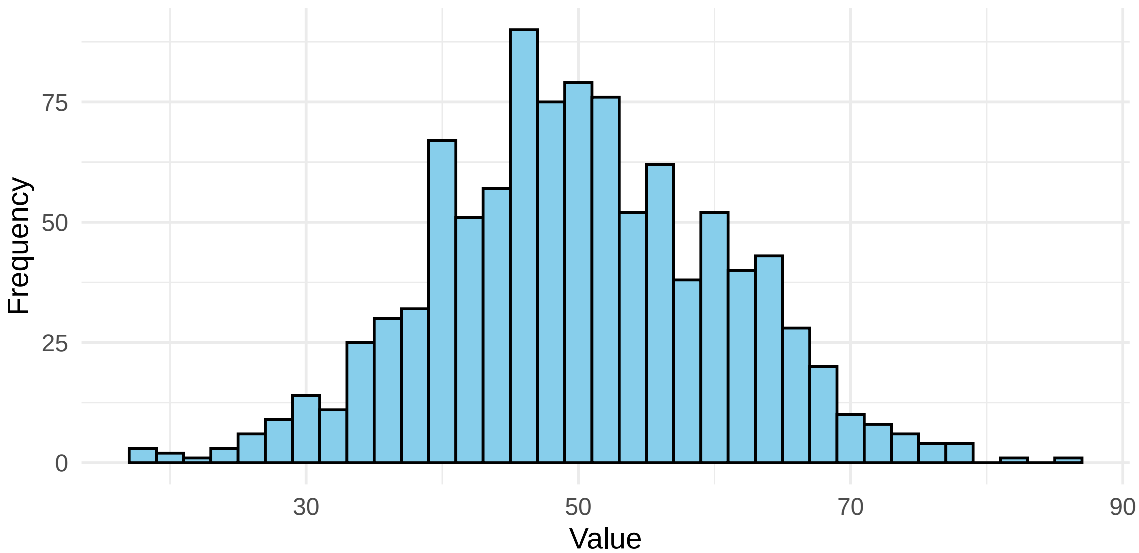
Discuss: Come up with 2 additional examples of variable that could be well-represented with a bar chart

3.5 Frequency Histograms

- A frequency histogram, at first glance, looks much like a __________ plot, as described prior.
- However, rather than use individual discrete points or labels, histograms will __________ values by a defined bin width, and count the frequencies of values within that bin
  - All the intervals will be the same __________, and we choose that width somewhat arbitrarily
  - But, its worth noting that the bin interval can have a __________ impact on the overarching interpretation

> **Important**
>
> Smaller bin size is not always better! Especially in data that is more spread out.

Figure 4: Example of a Histogram (Bin width = 2)

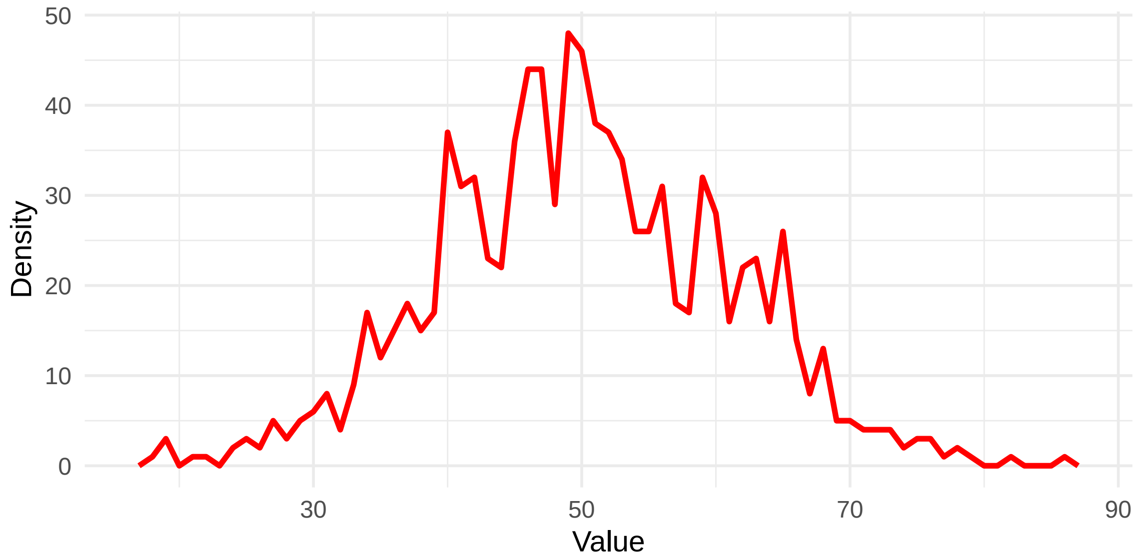


3.6 Frequency Polygons

> **Important**
>
> Frequency polygons look remarkably similar to line graphs, but start from a different perspective on understanding frequency.

- **Frequency polygons** can be though of as a line plot that that travels through the ____________ of histogram bars
  - So frequency polygons still use those same ____________ widths as with histograms, but don't show that on the graph itself (unless overlaid)
  - This makes them more appropriate for dealing with continuous data

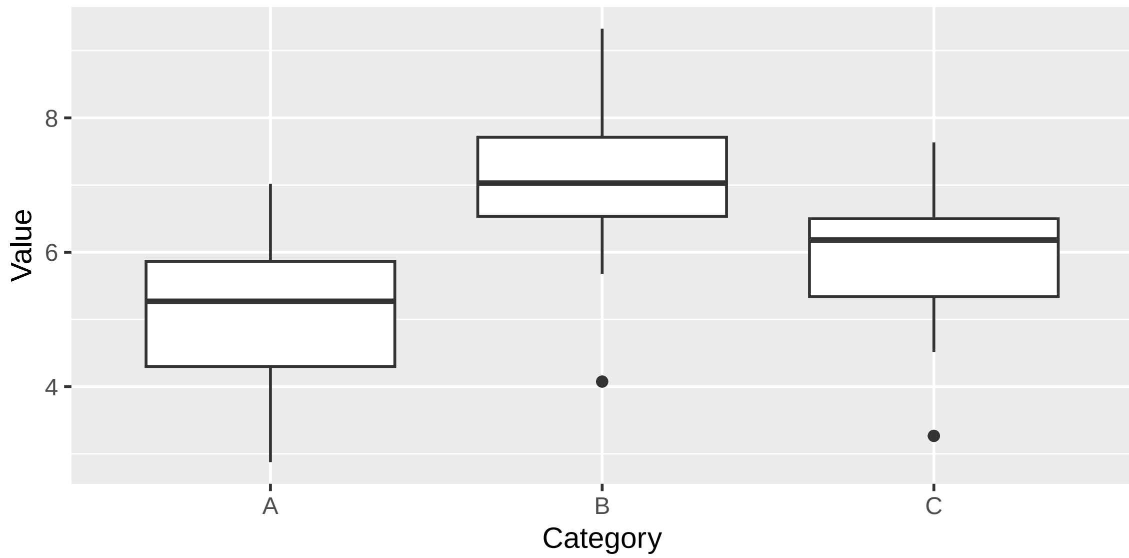
Figure 5: Frequency polygon



3.7 Box Plots

- Boxplots are useful for representing information about the quartiles, percentiles, and ________________ all in a single plot
  - Also called **box-and-whisker** plots (may be the preferred name for the cat-lovers like myself)
- The ________________ of a box plot represents a median, edges of the box represent Q_1 and Q_3 (i.e., the box is the IQR), the whiskers *usually* extend to the farthest values

Figure 6: Boxplot example



3.8 Practicing Choosing Descriptives and Plots

- Not all plots and descriptive statistics provide the ________________ information - try the following questions for practice

❓ I want to graphically examine the median and IQR of SAT scores in a local school district, how could I do this?

- (●) A) Boxplot
- ( ) B) Frequency histogram
- ( ) C) Kurtosis
- ( ) D) Arithmetic Mean

Explanation:

❓ I want to know the numeric value most commonly appearing in a dataset, how could I find this?

- (●) A) Mean
- ( ) B) Median
- ( ) C) Boxplot
- ( ) D) Mode

Explanation:

❓ I want visually compare my variable of student wellbeing ratings (continuous) against a normal distribution, how could I best examine this?

- (●) A) Boxplot
- ( ) B) Bar plot
- ( ) C) Histogram
- ( ) D) Mean

Explanation:

3.9 Aside about Lying in Statistics

- With all of these different methods (any many more!) for ________ representing data, we run into an issue: accidental or purposeful misleading with graphs
  - *"There are three kinds of lies: lies, damned lies, and statistics"* - Mark Twain (kind of, its complicated who exactly came up with this)
  - While often times taken out of context, this statement does ring true sometimes - we owe it to our readers to be careful

❗ Important

One of the best ways to ensure that data is represented fairly is to try to represent it via multiple different methods - these different methods will help us see slightly different dimensions and information about the distribution.

- Aim for *consistency and clarity in graphing*, avoiding axis points, confusingly similar colors, and making sure enough information is included to understand the scale of the plot.
  - Easy tip: see if you can show the plot to someone not knowledgeable in the work and see if it makes sense to them

4 Conclusion

4.1 Recap

- Understanding our data first starts with describing it, and we can accomplish that both through informative statistics and graph
- Measures of central tendency and dispersion can succinctly describe the center of data, and how spread out it is, respectively. They can be a useful way to quickly describe data in numbers.
- The various graphs show slightly different information, and multiple options may be used simultaneously to more thoroughly show the characteristics of the data
- The statistics we calculate on our sample are meant to accurately estimate the parameters of the population distribution, but that only works if our sample is representative and our data unbiased!
- Outliers and discussion on skew/kurtosis will return later when we are talking about their impact on inferential statistics, but don't worry too much about that yet!

Key Terms

B

Bar graphs Show frequencies of values of categorical/nominal variables 21
bimodal When a distribution has two peaks 15
bin width The range of each 'bar' in a frequency histogram 22
Boxplots A graphical representation of a continuous variable that shows the median, 25th percentile, and 75th percentile 24

C

Cumulative frequency A count of the number of cases that fall below a certain score 6
Cumulative relative frequency The percentage of scores that fall below a certain score 6

D

deviation How far away a point is away from the mean of the data 10

E

empirical rule A way to understand the percent of values at a certain number of standard deviation in a normal distribution 17

F

frequency histogram A graphical representation of a continuous variable using bin width ranges to count frequencies falling withing certain ranges 22

I

interquartile range (IQR) is the 75th percentile (upper quartile) minus the 25th percentile (lower quartile). It is a measure of spread. It is the width of the interval that contains the middle 50 percent of the data. 11

K

Kurtosis A description of how 'peaked' a distribution is 15

L

leptokurtic When a distribution has scores clustered towards the center 15

M

mean The arithmetic average of the data 8
Measures of central tendency statistics that summarize the typical, central, or average scores in a distribution 7
Measures of dispersion statistics that summarize how data is spread out across a distribution 10
median The value in the data in which half of values in the data fall below it, when ordered from smallest to largest 8
Modality A description of how many 'peaks' a distribution has 15
mode The most commonly occurring value in a variable 7

N

negative/left skew When a distribution has most values clustered towards the high end, and a tail out to the left side of the frequency histogram 14

normal distribution A distribution of a continuous variable that is unimodal, symmetrical, bell-shaped, and the mean, median, and mode are equal 17

O

outlier A point which is especially far away from the mean, usually by having an especially large z-score 13

P

Percentiles Points in the data that divide the data into 100 equal-sized amounts 3
platykurtic When a distribution has scores clustered in the tails 15
positive/right skew When a distribution has most values clustered towards the low end, and a tail out to the right side of the frequency histogram 14

Q

Quartiles Points in the data that divide the data into 4 equal-sized amounts 3

R

range The difference between the largest and smallest values in the data 11
Relative frequency The percent of total cases that fall in a certain category or at a certain point 6

S

sampling variation The natural difference in statistics from sample to sample 17
Simple frequency A count of the number of cases that fall in a certain category or at a certain point 6
Skewness A description of how a distribution departs from symmetry or has a 'tail' 14
standard deviation A measure of the average distance away from the mean that a point is in a distribution 10

U

unimodal When a distribution only has one peak 15

V

variance The average squared deviation scores from the mean 10

Z

z-score A statistics of how far away a specified point is away from the mean of the data, in terms of standard deviation 13

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms